

Machine Learning for Customer Churn Prediction

Nelson

Javier

Nicolas Castro Rivera

Abstract

Customer Churn is a problem about customer discontinue a company's services or products. Machine learning techniques can be applied with purpose of predict if a customer has probability of be churn, this information can help the company to take decisions for not lost clients. This work search explore different machine learning techniques to compare and get the best approach to solve this problem.

1 Introduction

Customer Churn refers to clients that potentially can discontinue a service or a product of a company. The main context for this project is the telecommunication

This should be a section of about one page. Here, the context of the problem should be fully described. Also, previous solutions to the same problem should be referenced. Most of the bibliography is cited here. While referencing previous solutions, you can also mention computational techniques used, which provides a complete reference for the reader. Usually, this section does not include images or tables.

2 Preprocessing data

Here you should describe the design of your solution, your technical decisions, and why you believe your solution is appropriate. Some general images are recommended to enhance your explanation, and this section is expected to be 1 to 2 pages long. It is not recommended to include code here, but if you are presenting a complex algorithm, you may add it. Remember to write full paragraphs, with one idea per paragraph, and avoid using itemized lists. Aim for clarity and readability. Include enough detail for reproducibility. Briefly mention any limitations or assumptions in your approach.

2.1 EDA

Dataset has the next features:

Feature	Description
customerID	Unique value that identify a customer
gender	Determines whether customer is Male or Female
SeniorCitizen	Determines whether a person is citizen or not
Partner	Determines whether person has partner
Dependents	Determines whether customer has a dependent
tenure	Number of months has stayed with the company
PhoneService	Whether customer has phone service
MultipleLines	Whether the customer has multiple lines, or one, or does not have phone service
InternetService	Internet service provider (Fiber optic, DSL and other)
OnlineSecurity	Whether customer has online security, or not, or does not have internet service
OnlineBackup	Whether customer has online backup, or not, or does not have internet service
DeviceProtection	Whether customer has device protection, or not, or does not have internet service
TechSupport	Whether customer has tech support, or not, or does not have internet service
StreamingTV	Whether customer has streaming tv, or not, or does not have internet service
StreamingMovies	Whether customer has streaming movies, or not, or does not have internet service
Contract	Contract term duration (Month-to-month, One year, Two year)
PaperlessBilling	Whether customer has paperless billing or not
PaymentMethod	Payment method (Electronic check, Mailed check, Bank transfer (automatic), Credit card)
MonthlyCharges	Value paid by customer monthly
TotalCharges	Total value paid by customer
Churn	Whether the customer churned or not, stopped using the service last month.

Table 1: Dataset features description

2.2 Subsection 2

Subsection example 1

2.3 Subsection 3

Subsection example 2

3 Results and Discussion

Explain the experiments you performed to demonstrate that your solution works. Summarize your unit tests in a paragraph, mentioning their philosophy, quantity, and results. Also, discuss any integration and acceptance tests to address software quality and user expectations. If possible, compare your results with other solutions using metrics. Use tables to summarize test definitions and results, and charts (such as box-plots or time series) for comparisons, depending on your project. Highlight the most important findings. Every figure and table must have a descriptive caption and be referenced in the text.

4 Conclusion

Write a couple of paragraphs to summarize your work, results, and achievements. Clearly state why your work was successful and how it helped to solve the problem. End with suggestions for future work or open questions that remain

References

- [1] Rummery, G. A., & Niranjan, M. (1994). On-line Q-learning using connectionist systems. *Technical Report*, Cambridge University Engineering Department.